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Office Contact Information
Department of Finance
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Education

Ph.D. in Economics, University of Chicago
D.Sc. in Technology (mathematics), Aalto University
M.S. in Mathematics, New York University
M.Sc. in Mathematics, University of Oulu

References:

Professor Tarun Ramadorai
Imperial College London Business School
t.ramadorai@imperial.ac.uk

Professor Veronica Guerrieri
Univ. of Chicago Booth School of Business
veronica.guerrieri@chicagobooth.edu

Professor Joseph Vavra
Univ. of Chicago Booth School of Business
Reference contact: abarbosa@uchicago.edu

Research Fields

Primary fields: Real Estate Finance, Macro Finance
Secondary fields: Household Finance, Macroeconomics

Teaching Experience

Instructor / Module leader

Autumn, 2022	Investments and Portfolio Management, Imperial College London
Spring, 2021	Economic Policy Analysis (advanced undergraduate macro), U of Chicago
Autumn, 2019	Economic Policy Analysis, University of Chicago
Summer, 2016	Differential Equations, University of Jyväskylä

Teaching Assistant

Autumn, 2023	Principles of Finance, Imperial College London
Autumn, 2019	Macroeconomics (MBA), Chicago Booth
Summer, 2019	Macroeconomics (MBA), Chicago Booth
Spring, 2019	Economic Policy Analysis, University of Chicago
Winter, 2019	Investments (MBA), Chicago Booth
Autumn, 2018	Macroeconomics (MBA), Chicago Booth
Spring, 2010	Basic Course in Mathematics L4 (Honors PDEs), Aalto University
Spring, 2009	Basic Course in Mathematics L4, Helsinki University of Technology
Autumn, 2008	Calculus I, New York University

Employment

2022 – 2025	Imperial College Business School, postdoctoral research associate (finance)
2015 – 2016	University of Jyväskylä, postdoctoral researcher (mathematics)
2012 – 2015	University of Helsinki, Academy of Finland postdoctoral researcher (mathematics)
2012	University of Oulu, postdoctoral researcher (mathematics)
2010 – 2012	The Boston Consulting Group, Associate

Honors, Scholarships, and Fellowships

2021	Yrjö Jahnsson Foundation Graduate Grant
2020	Data Acquisition Grant, University of Chicago Dept. of Economics
2017	CRSP Summer Research Grant, Booth School of Business
2016 – 2022	Social Science Graduate Fellowship, University of Chicago
2014 – 2016	Väisälä Foundation Travel Grants
2012 – 2015	Academy of Finland postdoctoral research fellowship in mathematics
2007 – 2009	Fulbright Foundation Scholarship
2007	Alfred Kordelin foundation Graduate Scholarship
2006	Ernst Lindelöf prize (master's thesis award), The Finnish Mathematical Society

Professional Activities

<u>Conference and Seminar Presentations (economics and finance):</u>

2024	Urban Economics Association European Meeting, Normac Junior Macro Conference, Helsinki Finance Summit (scheduled), Bank of Finland (scheduled)
2021	Young Economist Symposium (Princeton), 16th Annual Economics Graduate Student Conference (WUSTL), UChicago Joint Program and Friends Conference

<u>Referee (economics and finance):</u>
Review of Financial Studies

<u>Referee (mathematics):</u>
Journal de Mathématiques Pures et Appliquées, Nonlinear Analysis: Theory, Methods & Applications, Manuscripta mathematica, Annales Academiæ Scientiarum Fennicæ Mathematica, Journal of Mathematical Analysis and Applications, Applied Mathematics Letters

Language and Computer Skills

<u>Computer Skills:</u> R, Matlab, Dynare, Mathematica
<u>Languages:</u> English (fluent), Finnish (native)

Working Papers:

"Entrepreneurs and Housing Boom-Bust Dynamics" (Job Market Paper, draft available soon)

Abstract: Housing boom-bust periods are often preceded by financial liberalizations. Nevertheless, credit conditions' role in explaining macroeconomic dynamics during housing boom-bust episodes is sharply debated. Building on a growing literature connecting entrepreneurial financing to housing market dynamics, I consider how entrepreneurial credit constraints affect the macroeconomy in a dynamic general equilibrium model with occasionally binding collateral credit constraints. A credit relaxation lifts productive entrepreneurs off their credit constraints, generating a sustained economic boom, explaining many of the market dynamics observed during the U.S. housing boom of the early 2000s, incl. a substantial and disproportionate increase in house prices (relative to other asset prices), decreasing interest rates during a sustained boom, a significant consumption boom, and an eventual endogenous bust resembling credit crunch dynamics, with decreasing output and a sudden drop in interest rates, but without an explicit additional credit shock.

"Behavioral Lock-in: Aggregate Implications of Reference Dependence in the Housing Market" with Cristian Badarinza, Tarun Ramadorai, and Jagdish Tripathy

Abstract: We show both empirically and theoretically that households' attachment to nominal anchors in the housing market creates aggregate nominal rigidities, with material economic consequences. A new statistic, the prevalence of "paper losses" in the stock of residential properties, is remarkably effective at explaining cross-regional variation in a range of important market outcomes. We setup and structurally estimate a dynamic search and matching model with reference-dependent agents, rich heterogeneity, and realistic financial constraints to rationalize the data and study housing-market fiscal policy. Behavioral frictions dampen the effects of transactions taxes, and increase the revenue-maximizing level of the ongoing property tax rate.

"Entrepreneurs, Collateral Channel, and the House Price Volatility Puzzle"

Abstract: This paper proposes a new theoretical mechanism for explaining how a relaxation of entrepreneurial collateral constraints on business capital can increase house prices disproportionately. Such a credit relaxation enables productive entrepreneurs to purchase business capital, triggering capital reallocation and increasing aggregate productivity. Extending the Evans and Jovanovic (1989) model of entrepreneurial financing with housing markets, I demonstrate how this may produce a housing boom through general equilibrium effects. Empirical analysis leveraging Community Reinvestment Act (CRA) business loan data suggests that the mechanism can plausibly explain a notable portion of the pre-crisis increase in U.S. house prices during the early 2000s.

"Talent allocation, aggregate productivity, and income inequality: Evidence from Finland"

Abstract: This paper studies the impact of occupational talent allocation on aggregate productivity and income inequality. I use Finnish administrative microdata to estimate a Roy model of occupational choice with unobservable skill and preference heterogeneity. I show that workers' sorting behavior changes have not driven aggregate productivity growth in the recent past. Holding skill distribution fixed, potential future gains for aggregate productivity by improving sorting are also limited. However, different time trends on occupational sorting patterns explain up to 40% of relative wage growth in certain occupations during 1995 – 2005. In particular, differential occupational sorting between genders explains 3.5 percentage points, or 16%, of the gender earnings gap. When accompanied by changes in the skill distribution, workers' occupational sorting behavior also matters for aggregate output. Removing gender differences in skills, in particular, would lead to a 28% higher GDP effect than complete gender equalization with identical sorting patterns. I augment this analysis by leveraging the staggered implementation of the *Finnish Comprehensive School Reform* to discipline another counterfactual exercise. I use the model to decompose the reform effect into its skill and sorting components. I show that the reform's differential impact on women increased aggregate productivity by one percent, half of which was due to the sorting channel.

Selected publications (mathematics)

On the parabolic Harnack inequality for non-local diffusion equations (with D. Dier, J. Kemppainen, and R. Zacher). *Math Z.* 295 (2020), 1751–1769.

Boundary regularity for the porous medium equation (with A. Björn, J. Björn, and U. Gianazza). *Arch. Rational Mech. Anal.*, 230 (2018), no. 2, 493-538.

On the interior regularity of weak solutions to the 2-D incompressible Euler equations (with J.M Urbano). *Calc. Var. Partial Differential Equations*, 56, 126 (2017).

Representation of solutions and large-time behavior for fully nonlocal diffusion equations (with J. Kemppainen and R. Zacher). *J. Differential Equations*, 263 (2017), no. 1, 149-201.

Everywhere differentiability of viscosity solutions to a class of Aronsson's equations (with C. Wang and Y. Zhou). *Ann. Inst. H. Poincaré Anal. Non Linéaire*, 34 (2017), no. 1, 119-138.

Decay estimates for time-fractional and other non-local in time subdiffusion equations in R^d (with J. Kemppainen, V. Vergara, and R. Zacher). *Math. Ann.*, 366 (2016), no. 3-4, 941-979.

Hölder continuity for Trudinger's equation in measure spaces (with T. Kuusi, R. Laleoglu, and J.M. Urbano). *Calc. Var. Partial Differential Equations,* 45 (2012), no. 1-2, 193-229.

Local Hölder continuity for doubly nonlinear parabolic equations (with T. Kuusi and J.M. Urbano). *Indiana Univ. Math J.,* 61 (2012), no. 1, 399-430.

Obstacle problem for nonlinear parabolic equations (with R. Korte and T. Kuusi). *J. Differential Equations,* 246 (2009), no. 9, 3668--3680.